Cameron Beebe, Ph.D.

Agent-friendly: cameronbeebe.github.io/cv.md

Executive Summary

Interdisciplinary AI researcher and mathematical philosopher. Ph.D. (2015-2021) on cybernetics and AGI, deep learning internship at Cray in 2019. Recent 2026 work on a multi-agent LLM harness at Appy.AI (prompting, context management, orchestration, agent skills, and theory-driven evaluation). Through my LLC the SciPhi Initiative, I consult and help build agent-assisted business workflows, lead community outreach on AI and society, and push LLMs in experiments and projects. I drive towards conceptual clarity, am familiar with formal methods, and have been thinking about machine epistemology for a decade.

Experience

Dec. 2025–Oct. 2026 Appy.AI. AI Product Engineer on a multi-agent harness for Violet: a Chief of Staff that routes to specialist agents with scoped context for real workflows.

Shipped a harness eval dashboard comparing batches of executions across configurations. Decided against an external eval framework (e.g. Braintrust) in favor of in-house solutions. Drove qualitative theory-led improvements at low volume. Matched context to harness mechanics (affordance and ergonomics). Orchestration, steering, context management, skills.

2021– The SciPhi Initiative. My LLC for AI research, consulting, and community outreach. Researching capabilities and limitations of LLM harnesses and agents; projects below.

Oct. 2026: Building an agent harness and custom dashboard for High Performance Earthworks.

Leading discussion sessions on AI and society at Nederland Philosophy Club (co-organizer).

2023–2025 Colorado Outdoors. Snowboard instructor at Eldora Mountain Resort. Heavy equipment operator at High Performance Earthworks.

2015–2021 PhD at GSN (LMU Munich). AI, cybernetics, and philosophy of science in a structured Ph.D. program at the Graduate School of Systemic Neurosciences.

Dissertation on Knowledge Transfer in Cognitive Systems Theory. Transfer learning in artificial neural networks as effective transfer of control, related to model-based knowledge transfer in cognition. Supported Ashby's structural view of systems (cybernetics), and a model-based view of computation for diverse computational devices. Took a quantum information course from the physics department, and critiqued quantum-like models of cognition.

Affiliated with the Munich Center for Mathematical Philosophy and the Research Center for Neurophilosophy and Ethics of Neurosciences during this time. Presented work internationally.

2019 Deep Learning Intern at Cray, Inc. Worked for Cray's Distributed Deep Learning Plugin team. Applied, tested, and experimented with the Plugin across multiple GPUs (gradient averaging using MPI), on a variety of architectures including: LSTM, CNN, RL/DQN, CapsNet. Worked mainly with Keras, scheduled jobs via Slurm, managed ML/DL packages and versions with Anaconda and pip, and committed code to internal git repos. (**Python, Jupyter, Slurm, Linux, Keras, Anaconda, Bash, Git**).

Selected Projects (more on)

Intent Controller Harness mod (hermes) to make agents better follow intent.

`diaspy`**:** DSPy experiments with dialectical prompt-rewrite proposals inside GEPA.

`fungi_fortress` / `fungi_chromatic`**:** Generative game harness experiments. LLMs act through typed interfaces, **Jev** classification, flashing ROMs to cartridge.

`scipfs`**:** Agentic IPFS sharing.

Structural Idealism: Interpreting wave mechanics and quantum probabilities as ideal structural objects.

Transfer Learning: Intro tutorial with `keras`.

Modeling Memory in Signaling Games: Reinforcement learning with dropout.

Publications

Model-Based Computation. Natural Computing, Volume 17, Issue 2, p. 271-281. 2018.

Sequent Calculus Representations for Quantum Circuits. Electronic Proceedings in Theoretical Computer Science, volume 214, p. 3-15. 2016.

WIP Manuscripts (More on my ResearchGate.)

Artificial Neural Networks as Cybernetic Regulators. (Forthcoming)

Bayesian Confirmation from Analog Models. With Roland Poellinger.

Transfer Learning: The Logic of Discovery for Artificial General Intelligence

Decoherence and Survival: Why Incompatible Observables Should be Compatible.

Education

2021 Ph.D. from Graduate School of Systemic Neurosciences, LMU Munich.
Dissertation: *Knowledge Transfer in Cognitive Systems Theory*
Committee: Stephan Hartmann, Christian Leibold, Ulrike Hahn, Gregory Wheeler
Reviewer: Peter Asaro

2015 Masters in Logic and Philosophy of Science from MCMP, LMU Munich.
Supervisor: Karim Thébault, Reviewer: Michael Cuffaro
Thesis: *Fluid Mechanical Models in Physical and Computational Contexts*

Program in mathematical philosophy and philosophy of science including courses on: confirmation theory, formal epistemology, Bayesian networks, probability theory, logic, information theory, and agent based modeling (**Netlogo**).

Specialized in philosophy of physics.

2012 B.A. in Philosophy from Montana State University.
Thesis: *Uncertainty in Quantum Theory* Advisor: Prasanta Bandyopadhyay